

Week 1 Assignment

Repository Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone>

Dataset:

<https://www.kaggle.com/datasets/yanmaksy/big-startup-secses-fail-dataset-from-crunchbase>

Guidelines:

- 10-page max, 1.5-spaced paragraphs, 11-sized font, a professional font format. An unlimited number of appendix pages. Submitted as pdf within your GitHub repository

Sections:

1. Your problem statement

The idea of predicting whether a start up will fail, get acquired, or go public is a highly targeted problem within the prediction modeling sector. Most models that are derived from a multi-variate data background depend on a few main drivers. The first driver being the amount of capital a private company raised. In our case, this becomes misleading, total funding is relatively based off of survival percentage, since companies with more exposure to the funding process have more time to make adjustments. Our project focuses on a different approach leaning towards the question: Does the timing of funding predict outcomes better than the amount? Observing the exact amount of time a company took to accumulate funds in the first round, and how fast the overall pace was during the other rounds. We will then compare that model against a baseline that only considers total dollars raised.

The second part of our problem is predicting when the company fails. A startup that fails within a two year period is much different from one that survives until year 8; however, a normal classifier treats this the same. Companies that are still operating pose a slight problem since they have not necessarily succeeded or failed yet. In order to incorporate this, we must create a survival analysis which counts those companies as more of a censor rather than a failed company, not impacting risk and failure over time.

2. Articulation of value

The main value of this project is being able to provide timing features that are known early, where total funding is not. An investor that looks at an early stage company cannot predict how much that company will raise over its lifetime. However, investors can observe how long it took to get to the first round of funding and how quickly the next one followed. Depending on if those features predicted the outcome well, the model can pick up the exact point of precision the deal was made. A model built only on the lifetime of funding totals can only predict the success after the outcome.

The survival analysis piece adds a complex layer of value to our project. Instead of a single yes or no prediction layer, it will show when the risk is highest in a company's life span. This becomes very useful to funding managers, and investors on how to manage a portfolio and will show the paces of fundraising. The decision that come from the paces of fundraising will allow investors to identify in the early stages who needs support and when to decelerate funding.

3. Calculation of the potential economic value

- Document and footnote your assumptions
- Show how you calculated the value

Our model's economic value comes from letting a venture investor decide whether to make a follow-on investment in a startup before the outcome is known, using the funding-timing and velocity signals from section 1 instead of waiting on total funding raised.

To quantify this, we can assume a mid-size fund considers a follow-on investment in around 15 portfolio companies a year at their Series A and B round, investing an average of \$1.5 million per company, which is in line with typical reserve ratios and follow-on check sizes for funds of this size.¹ On average, three in four venture-backed startups fail to return their investors' capital,² and when a startup closes before reaching its next round, the money that has already been committed to it is almost completely lost, since a large share of VC-backed startups that go out of business return effectively nothing to investors.³

If our model correctly flags three additional high-risk startups a year that would have otherwise kept funding, then declining those follow-ons avoids around \$1.5 million in losses each, or roughly \$4.5 million each year across the portfolio.

This calculation assumes the fund loses the full follow-on check when a flagged startup shuts down, rather than only part of it, and well-funded startups that have failed back this assumption up. For example, Convoy raised its last round at a \$3.8 billion valuation in 2022 and shut down about a year later, losing that entire follow-on capital (Fortune, 2023). Additionally, Olive AI raised \$400 million at a \$4 billion valuation in 2021 and shut down two years later after multiple divestitures and layoffs (Healthcare Dive, 2023). Our estimate is in line with the kind of well-capitalized, high-growth startups this model is meant to flag.

*****Footnotes (need to re-format in the markdown, I just used numbers as placeholders here):**

1. Reserve ratios of roughly 40-60% and average follow-on checks around \$1.5M are typical for funds of this size (ValueAdd VC, 2026).
2. Gage, D. (2012). The venture capital secret: 3 out of 4 start-ups fail. *The Wall Street Journal*.

3. Booth, B. (2013). Correlation's fresh look at venture capital returns. *Forbes*. An analysis of 7,976 realized VC deals (2003-2012) found that 39% of them went completely out of business, returning effectively nothing to investors.

4. **Project plan: Build a 13-week plan. Identify the steps, identify the weeks, and what you will do in each step (you can look at the syllabus to build this plan)**

See external resource titled "Project Plan".

5. **Discuss the dataset: describe the dataset you've found, where you've found it from, and how will this dataset solve the problem statement?**

Our data comes from the Big Startup Success/Fail Dataset on Kaggle, which pulls company profile data from Crunchbase. It covers roughly 53,000 startups founded from the 1980s through 2015, and records each company's industry category, funding rounds, total funding raised, founding date, first and last funding date, location, and current status (operating, acquired, closed, or IPO) (Crunchbase, n.d.). The founding date and dated funding events for each company will allow us to calculate the time to first funding and the time between later rounds, which is what our problem statement focuses on. The status field also provides our outcome labels to sort each startup by closed, acquired, or IPO instead of only tracking whether a deal went through.

Most startups in this dataset haven't reached a final outcome yet, so the counts across categories aren't even. One analysis of it found that ~79% of companies still marked operating, ~9% closed, ~9% acquired, and less than 3% reaching an IPO (Ferraro, 2024). Those companies have not failed, they just haven't resolved yet, so we can treat them as censored observations instead of a fourth class to predict. That leaves a three-way classification between closed, acquired, and IPO for the outcome model, with the survival model built on top to capture when an outcome happens rather than if one does.

6. **Identify the type of modeling that will solve the problem you identified: supervised or unsupervised. If supervised, is this a classification or regression problem? If it is classification, is it binary or multi class? If it is unsupervised what type of unsupervised learning is it?**

The type of modeling that we will use to solve this problem falls under the umbrella of supervised learning, since we have a clearly defined target variable: three distinct company outcomes (failed, acquired, or went public). This clear outcome labeling rules out any unsupervised learning techniques, which would lack the guidance needed to train an effective model.

Within supervised learning, this is a multi class classification problem. While we could simplify the problem to binary classification (success vs failure), we are intentionally expanding the scope to capture three distinct outcomes because each scenario likely has different predictive drivers and value implications. A company that

gets acquired follows a different path than one that goes public or fails. Collapsing these into a binary prediction would muddy those critical differences.

The survival analysis component of the project is where it becomes particularly interesting. By layering survival analysis on top of our multi class classification framework, we address a severe limitation that standard classifiers don't: they treat all outcomes equally regardless of timing, and have difficulty with companies that are still operating with unknown features. Survival analysis lets us model when these outcomes are likely to occur across a company's lifecycle, not just whether or not they will. This lets us identify high risk windows for failure, and let us see when acquisition interest typically peaks or falls. For investors and portfolio managers, this means the model predicts when they should intervene for highest chances of success, or when they should stop the process overall.

Appendix Sources (APA format):

Booth, B. (2013, November 18). Correlation's fresh look at venture capital returns. *Forbes*. <https://www.forbes.com/sites/brucebooth/2013/11/18/correlations-fresh-look-at-venture-capital-returns/>

Crunchbase. (n.d.). *Big startup success/fail dataset from Crunchbase* [Data set]. Kaggle. <https://www.kaggle.com/datasets/yanmaksy/big-startup-success-fail-dataset-from-crunchbase>

Ferraro, B. (2024, January 22). *Is there a clear path to success for startups?* NYC Data Science Academy Blog. <https://nycdatascience.com/blog/student-works/is-there-a-clear-path-to-success-for-startups/>

Gage, D. (2012, September 20). The venture capital secret: 3 out of 4 start-ups fail. *The Wall Street Journal*. <https://www.wsj.com/articles/SB10000872396390443720204578004980476429190>

ValueAdd VC. (2026). *Reserve ratios in VC funds: How much to hold back for follow-ons in 2026*. <https://valueaddvc.com/blog/reserve-ratios-in-vc-funds-how-much-to-hold-back-for-follow-ons-in-2026>

Soper, S., Chapman, L., & Bloomberg. (2023, October 19). Convoy, a Bezos-backed trucking tech company valued at \$3.8 billion last year, is shutting down: "Today is your last day at the company." *Fortune*. <https://fortune.com/2023/10/19/convoy-trucking-tech-startup-shuts-down-jeff-bezos-investor>

Olsen, E. (2023, November 1). Health AI startup Olive to shut down. *Healthcare Dive*. <https://www.healthcaredive.com/news/olive-ai-shuts-down/698455/>